

ReLU, Leaky ReLU, ELU and Softmax

A single perceptron unless stated. All work by hand. Weight w only.

NAME	ROLL NO.	DATE
_____	_____	_____

PART 1 FORMULAS FROM LAST TIME

Keep these in front of you. You will not need to re-derive them.

The update rule

$$w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f(z) x \quad \text{where } z = wx + b$$

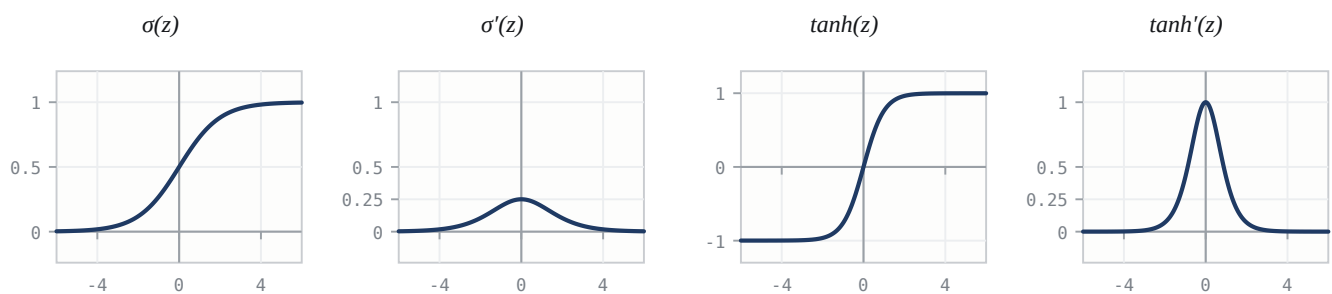
Sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad \sigma'(z) = \sigma(z)(1 - \sigma(z)) \quad \sigma'_{\text{max}} = 0.25$$

Tanh

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad \tanh'(z) = 1 - \tanh^2(z) \quad \tanh'_{\text{max}} = 1$$

FIGURE 1 – REFERENCE SHAPES FROM THE LAST WORKSHEET



The two derivative plots share a 0 to 1 vertical scale. Note how much lower σ' sits than $\tanh'$.

PART 2 SIGMOID'S REAL FAILURE: THE VANISHING GRADIENT

Last time we tested sigmoid at its **best** point, $z = 0$, where $\sigma' = 0.25$. Now test a case where the perceptron is almost as wrong as it can possibly be.

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f(z) x$

GIVEN $x = 1$ $w = 6$ $b = 0$ $y = 0$ $\alpha = 1$

USEFUL $e^{-6} = 0.00247875$

Step 1 (given). Weighted sum

$$z = wx + b = (6)(1) + 0 = 6$$

Step 2 (given). Activation

$$\hat{y} = \sigma(6) = \frac{1}{1 + e^{-6}} = \frac{1}{1.00247875} = 0.99752738$$

Step 3 (given). Error

$$\hat{y} - y = 0.99752738 - 0 = 0.99752738$$

Stop here before continuing. The sigmoid output must lie between 0 and 1, so the largest error possible is 1. Our error is 0.99752738, which is **99.75 % of the maximum possible mistake**. Write down the correction you expect, then work Steps 4 to 6.

I expect the weight to change by about _____

Step 4. Derivative

$$\sigma'(6) = \sigma(6)(1 - \sigma(6)) = \underline{\hspace{4cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \sigma'(z) x = \underline{\hspace{4cm}}$$

Step 6. Update

$$w_{\text{new}} = \underline{\hspace{4cm}}$$

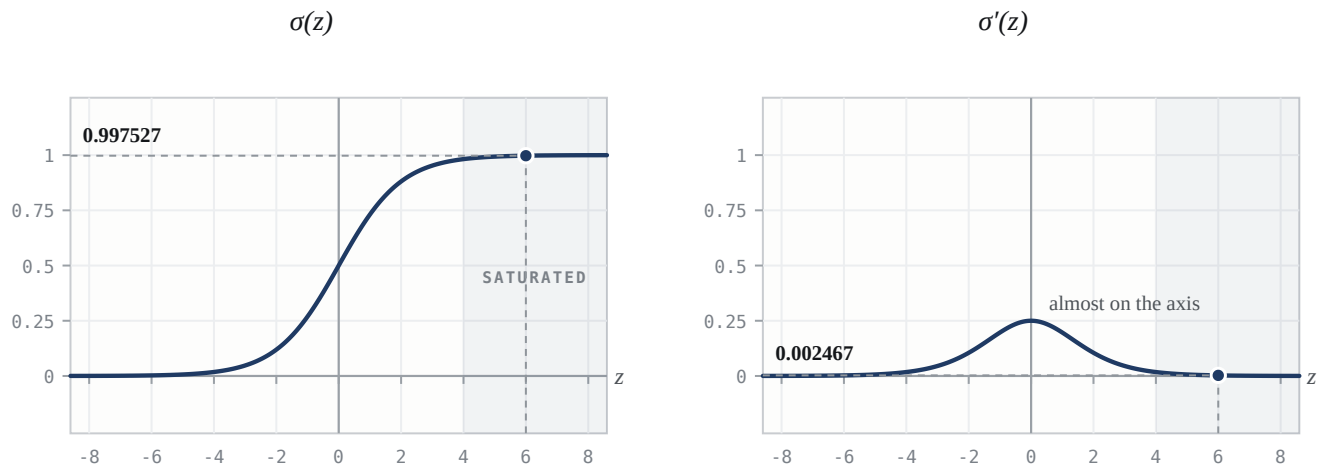
ANSWERS, FOR CHECKING ONLY

$1 - \sigma(6) = 0.00247262$ | $\sigma'(6) = 0.00246651$ | $\partial L / \partial w = 0.00246041$ | $w_{\text{new}} = 5.99753959$

2.1 The contradiction

	Value
Error	0.99752738 (almost the largest possible)
Step actually taken	0.00246041

FIGURE 2 – SIGMOID AT $z = 6$



Both panels share a 0 to 1 vertical scale. Shaded region: the saturated tail, where the curve is flat. Read the two marked values off the dashed guides.

Fill in.

The error is nearly the largest it can be, yet the step is only _____.

The factor responsible is _____, whose value here is _____.

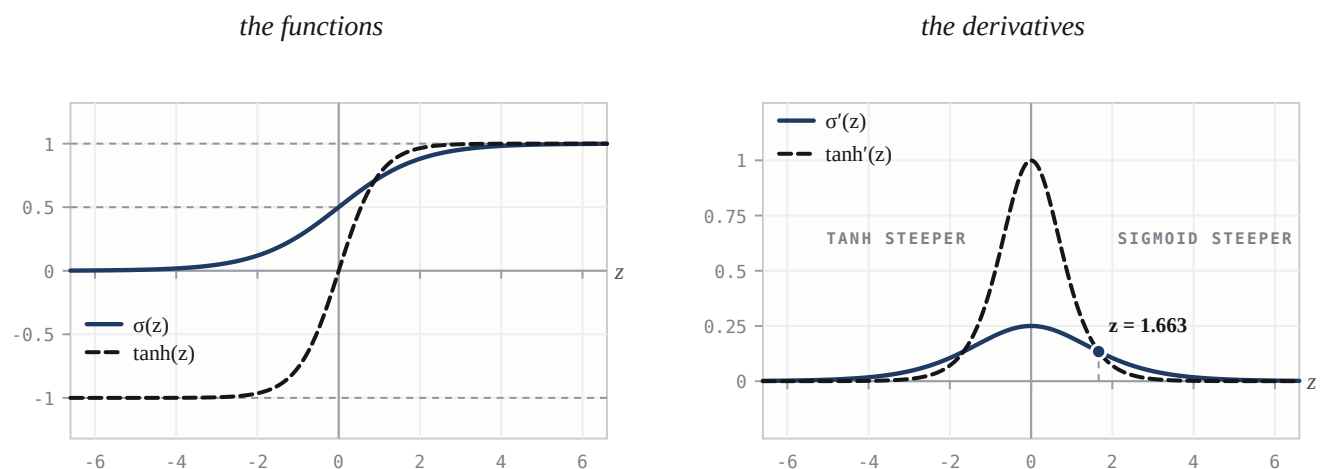
Reading Figure 2, the reason that factor is so small is

This is the vanishing gradient. When $|z|$ is large the sigmoid is flat, so $\sigma'(z) \approx 0$ and the perceptron cannot correct itself however wrong it is.

PART 3 TANH DOES NOT FIX IT

Tanh has a taller peak, 1.00 against 0.25, so it looks like the cure. Put both on the same axes and look carefully.

FIGURE 3 – SIGMOID AND TANH ON THE SAME AXES



Solid line: sigmoid. Dashed line: tanh. On the right the two derivative curves cross at $z = 1.663$. To the left of that point tanh is steeper; to the right it has already fallen below sigmoid and keeps falling faster.

Read the right-hand panel and fill in.

Near $z = 0$, the steeper derivative belongs to _____.

The two derivative curves cross at $z =$ _____.

Beyond the crossing point, the curve closer to zero belongs to _____.

At $z = 6$, both derivatives are effectively _____.

Tanh wins near the origin and loses badly once $|z|$ grows, because it flattens even faster than sigmoid. Both are S-shaped, both go flat, both vanish.

The S-shape itself is the problem. No rescaling of an S-curve will fix this.

PART 4 RELU

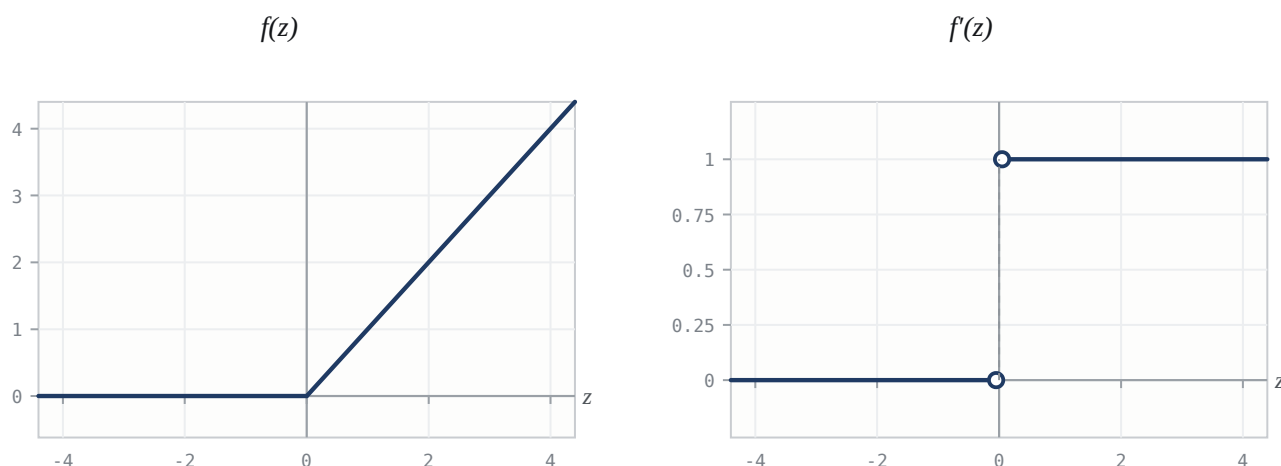
4.1 Definition

Stop bending an S. Use a function that never flattens on the positive side.

$$f(z) = \max(0, z) = \begin{cases} z & \text{if } z > 0 \\ 0 & \text{if } z \leq 0 \end{cases} \quad f'(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$

At $z = 0$ there is a corner, so $f'(0)$ does not exist. Libraries take it as 0.

FIGURE 4 – RELU AND ITS DERIVATIVE



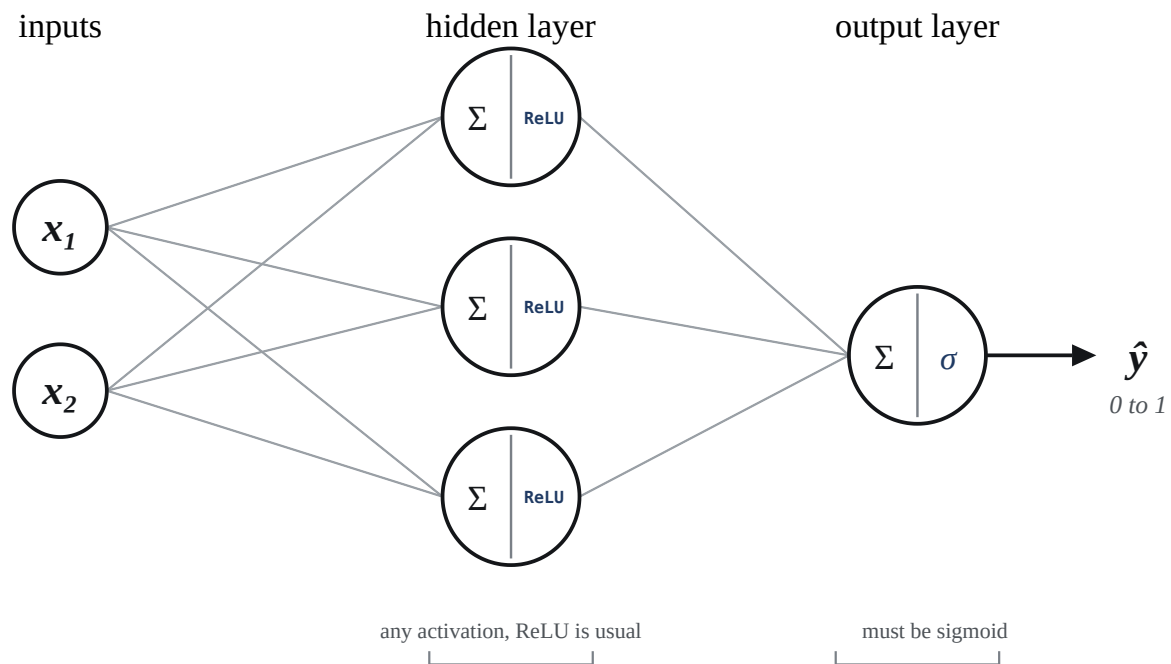
Note what is missing on the right: the derivative for $z > 0$ is the constant 1. There is no z in it, so it cannot shrink. $f'(6) = 1$ and $f'(600) = 1$.

4.2 Side question: can ReLU be the final activation?

Last worksheet we said an activation function exists to turn a raw score into an **answer**. ReLU does not do that. Its output is unbounded, so $\text{ReLU}(6) = 6$, which is not a decision.

So ReLU goes in the **middle** layers. The **final** layer must still squash the score into 0 to 1, and for that we use sigmoid.

FIGURE 5 — WHERE EACH ACTIVATION SITS IN A NETWORK



Each hidden and output unit is drawn split in two: the left half does the weighted sum, the right half applies the activation function.

One line. Why can the final layer not use ReLU?

4.3 The same numerical, with ReLU

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN, UNCHANGED FROM PART 2 $x = \mathbf{1}$ $w = \mathbf{6}$ $b = \mathbf{0}$ $y = \mathbf{0}$ $\alpha = \mathbf{1}$

Step 1 (given). Weighted sum

$$z = wx + b = (6)(1) + 0 = 6$$

Step 2 (given). Activation

$$\hat{y} = \max(0, 6) = 6$$

Step 3. Error

$$\hat{y} - y = \underline{\hspace{2cm}}$$

Step 4. Derivative

$$f'(6) = \underline{\hspace{2cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = \underline{\hspace{2cm}}$$

Step 6. Update

$$w_{\text{new}} = \underline{\hspace{2cm}}$$

Step 7. Put your new weight back through the perceptron and check it

$$z = \underline{\hspace{1cm}} \quad \hat{y} = \underline{\hspace{1cm}} \quad \hat{y} - y = \underline{\hspace{1cm}}$$

ANSWERS, FOR CHECKING ONLY

$$\hat{y} - y = 6 \quad | \quad f'(6) = 1 \quad | \quad \partial L / \partial w = 6 \quad | \quad w_{\text{new}} = 0 \quad | \quad \text{then } z = 0, \hat{y} = 0, \text{ error} = 0$$

The error becomes exactly zero. Solved in one step, on the same problem where sigmoid moved by 0.0025.

Activation	f'(6)	Step taken
Sigmoid	0.00246651	0.00246041
ReLU	1	6

$$\frac{\text{ReLU step}}{\text{sigmoid step}} = \frac{6}{0.00246041} \approx 2439$$

Honest note. Two things changed here, not one. The error grew from about 1 to 6 because ReLU is unbounded, and the derivative grew from 0.0025 to 1. The error grew about 6×; the derivative grew about 405×.

ReLU passed on the positive side. Now test the other half. Work every step yourself.

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN $x = 1$ $w = -3$ $b = 0$ $y = 1$ $\alpha = 1$

Step 1. Weighted sum

$$z = wx + b = \underline{\hspace{2cm}}$$

Step 2. Activation

$$\hat{y} = \max(0, z) = \underline{\hspace{2cm}}$$

Step 3. Error

$$\hat{y} - y = \underline{\hspace{2cm}}$$

Step 4. Derivative

$$f'(z) = \underline{\hspace{2cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = \underline{\hspace{2cm}}$$

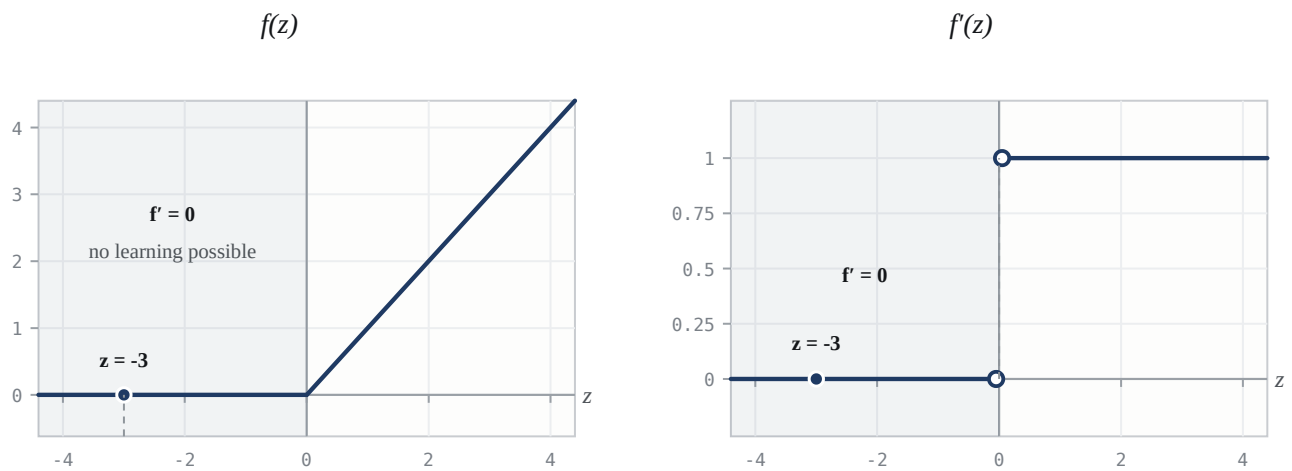
Step 6. Update

$$w_{\text{new}} = \underline{\hspace{2cm}}$$

ANSWERS, FOR CHECKING ONLY

$$z = -3 \quad | \quad \hat{y} = 0 \quad | \quad \hat{y} - y = -1 \quad | \quad f'(z) = 0 \quad | \quad \partial L / \partial w = 0 \quad | \quad w_{\text{new}} = -3$$

FIGURE 6 — RELU ON THE NEGATIVE SIDE



Shaded region: $z < 0$, where the derivative is exactly zero. A perceptron whose z lands here can never be corrected.

5.1 Two questions

Run the next update. What is z ? _____ What is the gradient? _____

Set $\alpha = 1,000,000$. What is the step size now? _____ Why?

This perceptron is not slow, it is dead. The gradient is exactly zero rather than merely small, and no learning rate can rescue a product containing zero.

PART 6 A TEMPTING WRONG ANSWER

Suggest a fix. How would you repair the negative side of ReLU?

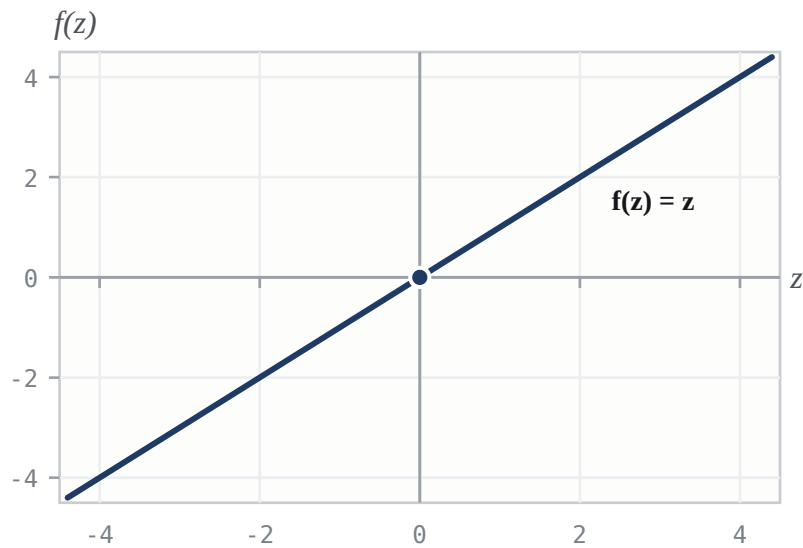
6.1 The proposal

Most classes arrive here, and it is worth taking seriously.

“ReLU works on the positive side because there $f(z) = z$, whose derivative is 1. So use $f(z) = z$ on the negative side too.”

That gives the **identity function**, $f(z) = z$ for every z , with $f'(z) = 1$ everywhere. The gradient is never zero, so no perceptron can ever die.

FIGURE 7 – THE IDENTITY FUNCTION



A single straight line through the origin. Its derivative is 1 for every z , positive or negative.

6.2 Why it cannot work

Substitute $f(z) = z$ into the perceptron:

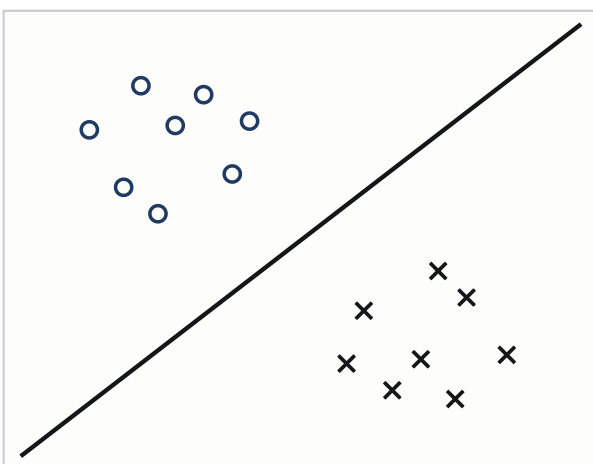
$$\hat{y} = f(z) = z = wx + b$$

The activation function has done **nothing**. It has passed the linear combination straight through, unchanged.

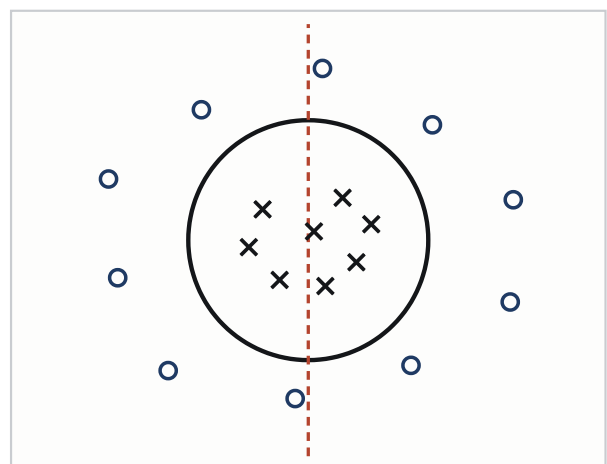
A straight line fed into a straight line is still a straight line, so however many layers we stack, the whole network stays one straight line. It can never bend.

FIGURE 8 – WHAT A STRAIGHT LINE CANNOT DO

linearly separable



not linearly separable



Left: one straight line separates the two groups. Right: no straight line can, shown by the dashed attempt. Only a curved boundary works.

Complete. The identity function fails because

What we need: a non-zero derivative on the negative side, as the identity gives us, but a shape that is not one straight line, so the network can still bend.

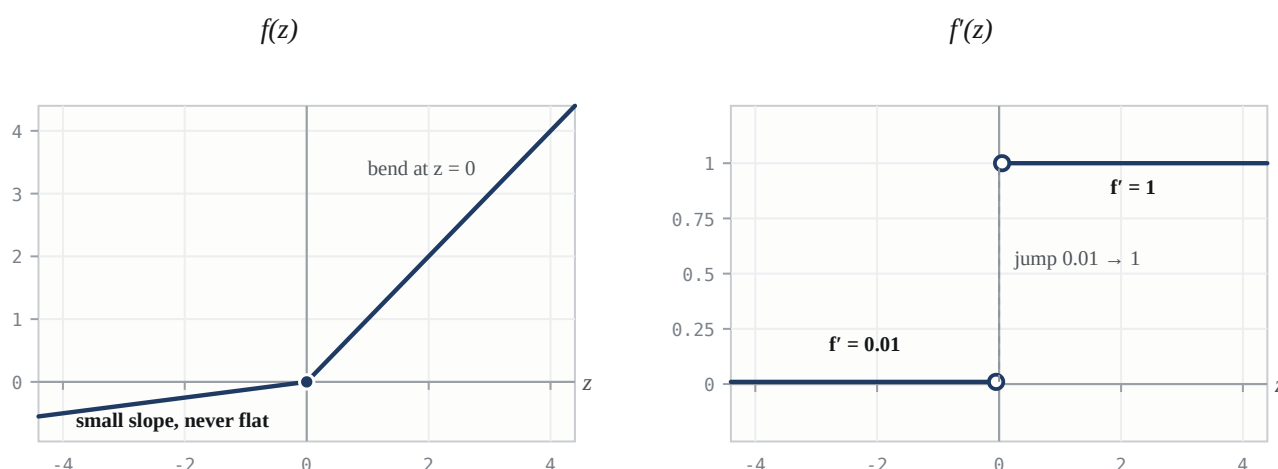
PART 7 LEAKY RELU

7.1 Definition

Keep the negative side a straight line, but give it a **different slope** from the positive side. That single change puts a bend at $z = 0$, so the function is no longer one straight line.

$$f(z) = \begin{cases} z & \text{if } z > 0 \\ 0.01z & \text{if } z \leq 0 \end{cases} \quad f'(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0.01 & \text{if } z < 0 \end{cases}$$

FIGURE 9 – LEAKY RELU AND ITS DERIVATIVE



The negative slope is drawn larger than 0.01 so that it is visible on paper. What matters is that it is not flat, and that it differs from the slope on the right, which puts a bend at $z = 0$.

Two checks.

Compare with the identity function. Leaky ReLU has slope _____ on the left and slope _____ on the right, so it _____ one straight line, and the network can still bend.

Compare with ReLU. On the negative side ReLU has derivative _____, while Leaky ReLU has _____, so no perceptron can die.

7.2 Numerical

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN, UNCHANGED FROM PART 5 $x = 1$ $w = -3$ $b = 0$ $y = 1$ $\alpha = 1$

Step 1. Weighted sum

$$z = wx + b = \underline{\hspace{2cm}}$$

Step 2. Activation, negative branch since $z \leq 0$

$$\hat{y} = 0.01z = \underline{\hspace{2cm}}$$

Step 3. Error

$$\hat{y} - y = \underline{\hspace{2cm}}$$

Step 4. Derivative

$$f'(z) = \underline{\hspace{2cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = \underline{\hspace{2cm}}$$

Step 6. Update

$$w_{\text{new}} = \underline{\hspace{2cm}}$$

ANSWERS, FOR CHECKING ONLY

$z = -3$ | $\hat{y} = -0.03$ | $\hat{y} - y = -1.03$ | $f'(z) = 0.01$ | $\partial L / \partial w = -0.0103$ | $w_{\text{new}} = -2.9897$

Activation	$f'(-3)$	Step taken	Alive?
ReLU	0	0	No
Leaky ReLU	0.01	0.0103	Yes

The perceptron is alive, but only just. It is crawling.

Two weaknesses of Leaky ReLU. Fill in from your working and from Figure 9.

The output on the negative side was only _____, which is barely different from zero, because the slope 0.01 is a small fixed constant chosen by hand.

At $z = 0$ the derivative jumps from _____ to _____, so the derivative is not continuous there.

PART 8 ELU

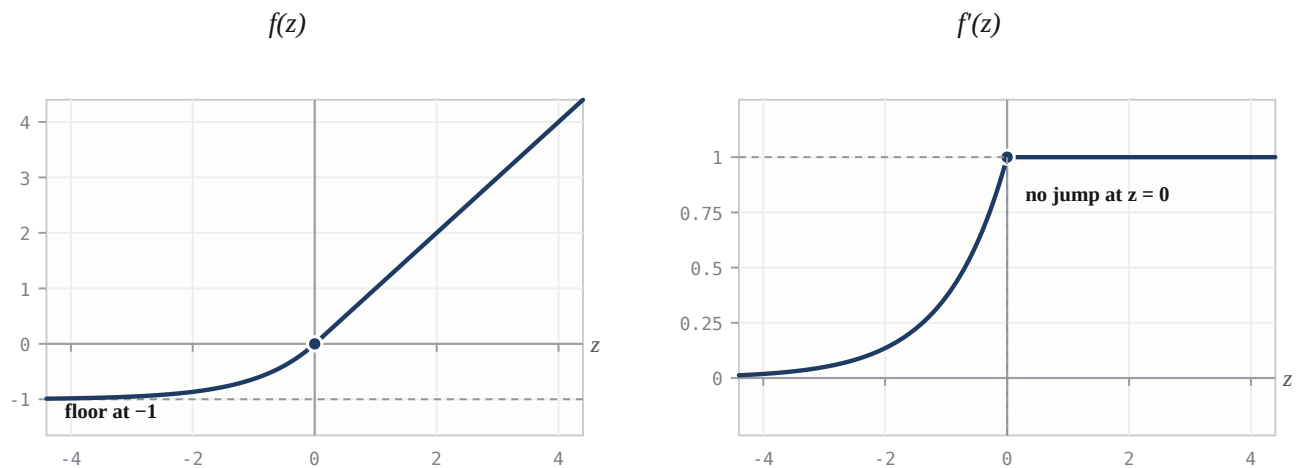
8.1 Definition

Keep $f(z) = z$ for positive z , exactly as ReLU. Replace the straight negative line with a **smooth curve** that flattens out to -1 .

$$f(z) = \begin{cases} z & \text{if } z > 0 \\ e^z - 1 & \text{if } z \leq 0 \end{cases} \quad f'(z) = \begin{cases} 1 & \text{if } z > 0 \\ e^z & \text{if } z < 0 \end{cases}$$

Recall that the derivative of e^z is e^z , and the constant -1 differentiates to zero. That is the whole derivation of the negative branch.

FIGURE 10 – ELU AND ITS DERIVATIVE



The negative side curves down to a floor at -1 instead of running off as a straight line. On the right, the derivative arrives at $z = 0$ from below at exactly 1, joining the positive branch with no jump.

Two checks.

As $z \rightarrow -\infty$, $e^z \rightarrow 0$, so $f(z) \rightarrow$ _____. The negative side is **bounded**, so it is not a straight line.

At $z = 0$: from the right, $f'(0^+) =$ ____; from the left, $f'(0^-) = e^0 =$ _____. They agree, so the derivative has _____, unlike Leaky ReLU.

8.2 Numerical

RULE $w_{\text{new}} = w_{\text{old}} - \alpha (\hat{y} - y) f'(z) x$

GIVEN, UNCHANGED FROM PART 5 $x = 1$ $w = -3$ $b = 0$ $y = 1$ $\alpha = 1$

USEFUL $e^{-3} = 0.049787$

Step 1. Weighted sum

$$z = wx + b = \underline{\hspace{2cm}}$$

Step 2. Activation, negative branch since $z \leq 0$

$$\hat{y} = e^z - 1 = \underline{\hspace{2cm}}$$

Step 3. Error

$$\hat{y} - y = \underline{\hspace{2cm}}$$

Step 4. Derivative

$$f'(z) = e^z = \underline{\hspace{2cm}}$$

Step 5. Gradient

$$\frac{\partial L}{\partial w} = \underline{\hspace{2cm}}$$

Step 6. Update

$$w_{\text{new}} = \underline{\hspace{2cm}}$$

ANSWERS, FOR CHECKING ONLY

$$z = -3 \quad | \quad \hat{y} = -0.950213 \quad | \quad \hat{y} - y = -1.950213 \quad | \quad f'(z) = 0.049787 \quad | \quad \partial L / \partial w = -0.097095 \quad |$$

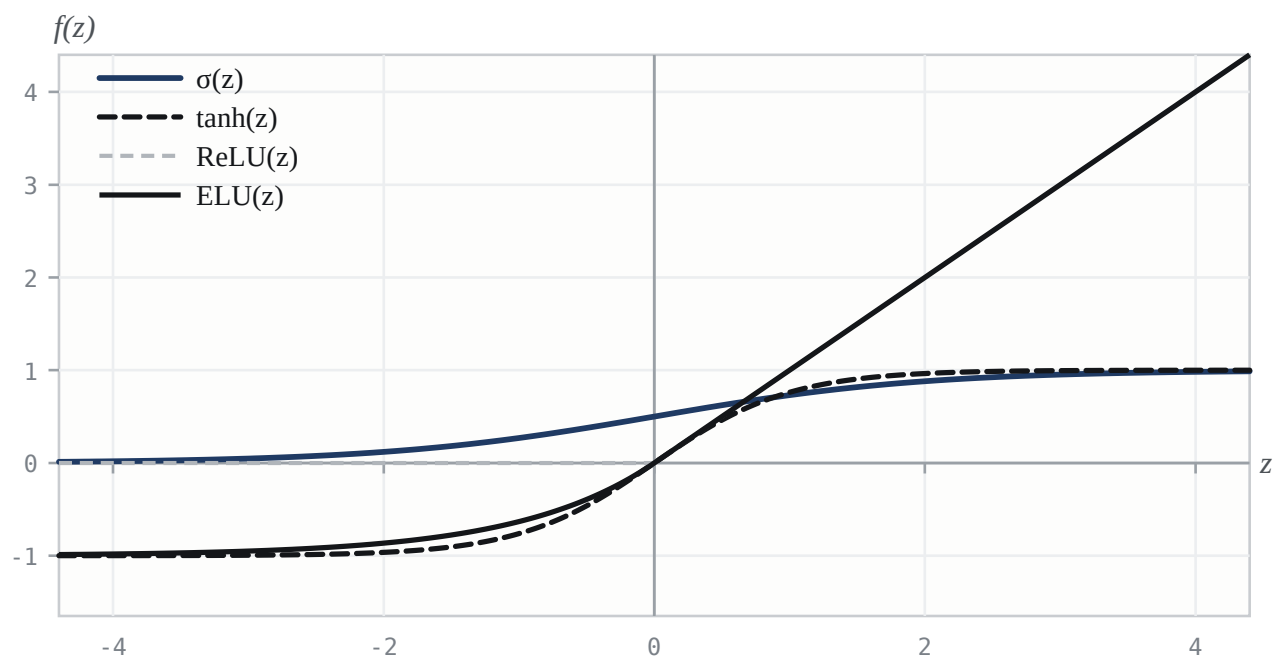
$$w_{\text{new}} = -2.902905$$
8.3 The verdict on the negative side

Activation	$\hat{y}$	$f'(-3)$	$\partial L / \partial w$	$w : -3 \rightarrow$	Verdict
ReLU	0	0	0	-3	Dead
Identity	-3	1	-4	1	Alive, but network becomes one straight line
Leaky ReLU	-0.03	0.01	-0.0103	-2.9897	Alive, crawling
ELU	-0.950213	0.049787	-0.097095	-2.902905	Alive, moving properly

$$\frac{\text{ELU step}}{\text{Leaky ReLU step}} = \frac{0.097095}{0.0103} \approx 9.4$$

Complete. ELU takes a step about _____ times larger than Leaky ReLU on the same problem, because its negative-side output is _____ rather than nearly zero, and its derivative there is _____ rather than a small fixed constant.

FIGURE 11 – ALL FOUR FUNCTIONS ON ONE SET OF AXES



Sigmoid and tanh flatten at both ends. ReLU and ELU keep rising on the right, which is why the vanishing gradient does not affect them there.

PART 9 COMPARISON

9.1 The mathematics

Function	$f(z)$	$f'(z)$	Output range	max f'
Threshold	1 if $z \geq 0$, else 0	0	$\{0, 1\}$	0
Sigmoid	$1 / (1 + e^{-z})$	$\sigma(1 - \sigma)$	$(0, 1)$	0.25
Tanh	$(e^z - e^{-z}) / (e^z + e^{-z})$	$1 - \tanh^2 z$	$(-1, 1)$	1
ReLU	$\max(0, z)$	1 or 0	$[0, \infty)$	1
Leaky ReLU	z or $0.01z$	1 or 0.01	$(-\infty, \infty)$	1
ELU	z or $e^z - 1$	1 or e^z	$(-1, \infty)$	1

9.2 In practice

Function	Advantage	Disadvantage	Typical real-world use
Threshold	Gives a clean yes/no answer	Derivative is 0, so it cannot be trained at all	Historical only, the original 1958 perceptron
Sigmoid	Output reads directly as a probability	Vanishing gradient for large $ z $; caps at 0.25 even at best	Final layer for binary classification: spam or not, disease or not; gates inside LSTMs
Tanh	Zero-centred; steeper than sigmoid near the origin	Saturates faster than sigmoid, so vanishing gradient is worse	Hidden states in RNNs and LSTMs; older shallow networks
ReLU	No vanishing gradient for $z > 0$; very cheap; trains deep networks	Dies permanently for $z < 0$; unbounded, so unusable in the final layer	Default hidden-layer choice in CNNs for image classification
Leaky ReLU	Never dies; as cheap as ReLU	Negative output is tiny; derivative jumps at $z = 0$; the slope 0.01 is chosen by hand	Hidden layers in GANs, where dead units destabilise training
ELU	Never dies; smooth derivative at $z = 0$; bounded below, so not linear	Needs an exponential, so slower than ReLU; derivative still small for very negative z	Hidden layers where too many ReLU units are dying
Softmax	Turns several scores into probabilities that sum to 1	Meaningful only across a whole layer, never for one unit alone	Final layer for multi-class classification

PART 10 BEYOND BINARY: SOFTMAX

10.1 The question

A sigmoid in the final layer returns one number between 0 and 1, which answers one question: yes or no.

Now suppose we must choose between three classes: cat, dog, bird.

One sigmoid cannot do this. Nor can three separate sigmoids, because each would answer independently and their outputs would not add up to 1. We need one function that takes all three scores together.

Complete. For a valid set of class probabilities, every output must lie between _____ and _____, and all outputs together must sum to _____.

10.2 Definition

For a final layer of n units with raw scores $z_1, z_2, \dots, z_n$:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

The exponential makes every value positive. Dividing by the total forces them to sum to 1.

10.3 Worked numerical

GIVEN, RAW SCORES $z_1 = 2.0$ $z_2 = 1.0$ $z_3 = 0.1$

Class 1 = cat, class 2 = dog, class 3 = bird.

Step 1. Exponentiate every score

$$e^{2.0} = 7.389056$$

$$e^{1.0} = 2.718282$$

$$e^{0.1} = 1.105171$$

Step 2. Add them

$$\sum_j e^{z_j} = 7.389056 + 2.718282 + 1.105171 = 11.212509$$

Step 3. Divide each exponential by that total

$$p_1 = \frac{7.389056}{11.212509} = 0.659001$$

$$p_2 = \frac{2.718282}{11.212509} = 0.242433$$

$$p_3 = \frac{1.105171}{11.212509} = 0.098566$$

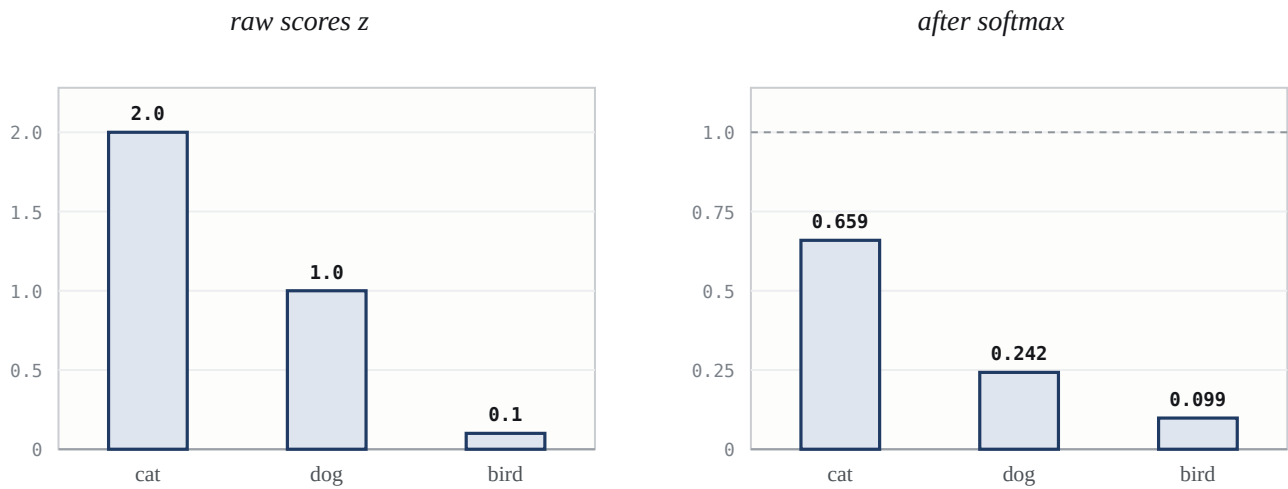
Step 4. Check the total

$$p_1 + p_2 + p_3 = 0.659001 + 0.242433 + 0.098566 = 1.000000 \quad \checkmark$$

Step 5. Read the decision

largest is $p_1 = 0.659001$, so the class is **cat**, at 65.9% confidence

FIGURE 12 – SOFTMAX TURNS SCORES INTO PROBABILITIES



Left: the raw scores, which can be any size. Right: the three probabilities, which add up to exactly 1. The order is unchanged.

10.4 Two observations

Order is preserved. The largest score gave the largest probability. Softmax does not change which class wins, it only converts the scores into probabilities.

Softmax with two classes is sigmoid. Take $n = 2$ with scores z and 0:

$$\text{softmax}(z) = \frac{e^z}{e^z + e^0} = \frac{e^z}{e^z + 1}$$

Divide numerator and denominator by e^z :

$$\frac{e^z}{e^z + 1} = \frac{1}{1 + \frac{1}{e^z}} = \frac{e^z}{e^z + 1}$$

Sigmoid is simply softmax for two classes. Binary classification is a special case, not a separate idea.

10.5 Closing summary

Task	Final-layer activation	Number of output units
Binary classification	Sigmoid	1
Multi-class classification	Softmax	one per class
Hidden layers	ReLU, or Leaky ReLU / ELU if units are dying	any

Final question. A network classifies images into 10 digits, 0 to 9. How many units are in the final layer, and which activation function do they use?

All numerical values in this worksheet are exact to six decimal places, except the two ratios marked with $\approx$. Every plot is drawn to scale, and all derivative plots share a vertical axis running from 0 to 1 so that peak heights may be compared directly.